

pyfs-matrix cheatsheet

Typical flow: inventory (new mesh) → plan (mandatory, no solver) → run (--resume to add points) → collect (cluster jobs) → post | Safe by default: storage, sync, restore, rebuild and mark-failed only PREVIEW until --apply. --workspace defaults to the current folder; --runs NAME picks another manifest than runs.json; a matrix name is found in the workspace root or inputs/matrices/.

1. PREPARE THE WORKSPACE AND THE MATRIX

inventory <geometry> write <stem>.boundaries.toml beside a saved simulation (.fsm, from its mesh block) or an OBJ (from its groups); names the solver actions saved in a .fsm; no executable needed

- overwrite rewrite an existing .fsm sidecar (an OBJ sidecar is never rewritten)
- clean remove the saved solver actions (26.120 in metres: also reset to a fresh import, meshes and BCs kept); copy kept as <file>.bak-<stamp>

upgrade <matrix> convert a matrix written under an older layout

- in-place rewrite the file (default: print to stdout)
- inputs DIR also move inputs/groups/e*.toml into pproc [groups]; needs --in-place

rename rename a 0.20.x workspace to the 0.21.0 point names (folders, scripts, manifest, plan)

- workspace DIR --dry-run print what would move, change nothing

convert <matrix> emit the equivalent native campaign.toml

- fs-exe PATH required: FlightStream executable (never guessed)
- name NAME campaign name (required here: no workspace)
- fs-version V build for rows with empty FS_BUILD, e.g. 26.124
- recipe CODE=MODULE:FUNCTION FS_SCRIPT code to recipe (repeatable)
- o, --output FILE write here instead of stdout

2. PLAN (MANDATORY BEFORE RUN, NO SOLVER)

plan <matrix> bind the matrix to inputs/, pre-flight every point, write the receipt run requires (pinned to the matrix digest; edit the matrix, plan again). Warns, never refuses, on saved actions in an unsteady geometry and on families a row lacks. Alias: inspect-setups

- workspace DIR campaign root with inputs/
- name NAME campaign name (default: workspace folder name)
- fs-version V fallback build for empty FS_BUILD cells
- fs-exe PATH executable override; mandatory for MANUAL rows (else FS_BUILD via inputs/executables.toml)
- workflow CODE=NAME FS_SCRIPT code to run type (repeatable)
- recipe CODE=MODULE:FUNCTION LEGACY recipe (repeatable)
- point-name TPL naming template; default P<sim>-<point name>; placeholders {point} {alpha} {beta} {mach} {advance_ratio} {sim} {campaign}
- ignore-missing-families [true|false] default true: a family the mesh lacks is skipped; false refuses the point
- no-ignore-missing-families same as false
- cost table mesh size, TEs, farfield layers, viscous coupling, steady/unsteady, iterations, CPUs and an EXPECTED wall time fitted on this workspace's runs
- inflow-fft for qsteady_rotor wheel points in a custom inflow: blade-frame harmonics, k_eff, n_max, suggested PASSAGE_POSITIONS (warns if too few)
- update-ids (--updateIDs) REWRITE this matrix so no POL repeats one used elsewhere; written before the plan runs
- accept-unregistered-build allow an installed build other than the registered one (your responsibility; recorded)
- setup-guidelines write inputs/setups/SETUP_GUIDELINES.md
- setup-standards write commented s9XX setup standards
- verbose full warnings (all detail also in logs/activity.log)

3. RUN

run <matrix> run every active point and write the sweep table. Needs a current plan. Linux with a submission profile submits to the cluster; Windows runs locally

- resume skip points already recorded (a grown matrix runs only new points); without it a recorded point refuses
- force-rerun POINT REDO a recorded point (by point name, run_id or steady JOB id; repeatable). Old records and outputs are archived, nothing deleted. Not with --resume
- force-rerun-all REDO every recorded point (prints the licence count first)
- sims SIM [SIM ...] with --force-rerun-all: only these simulations, space separated, e.g. --sims 2031 2032
- local run on THIS machine instead of submitting (Linux workstation or smoke test)
- progress-every N local unsteady progress every N steps (default 10; 0 = silent)
- sweep-csv FILE sweep table path (default post/<matrix>/campaign_sweep.csv)
- workflow CODE=NAME run type: steady, unsteady, unsteady_rotor, qsteady_rotor
- workspace, --name, --fs-version, --fs-exe, --recipe, --point-name, --ignore-missing-families, --no-ignore-missing-families, --accept-unregistered-build: as in plan
- pproc-warnings print grouped post-processing warnings
- verbose full warnings and one line per repeated item

4. COLLECT (CLUSTER OR INTERRUPTED RUNS)

collect for every SUBMITTED record, wait until its outputs are present and SETTLED, collect, assess, update the record, then post; HPC job_end_files present, no solver log: FAILED_EXECUTION

- watch keep sweeping until nothing is outstanding (default: one sweep)
- watch-interval S seconds between sweeps (default 60)
- interval S seconds between the two size/time observations (default 2)
- rounds N stop a watch after N sweeps
- sims IDS only these simulations, e.g. 2006,2007, and their post
- no-post collect only, do not rebuild products
- check-frozen REFUSE (not warn) averages touched by a frozen solve or an unread log block
- workspace, --runs NAME, --pproc-warnings, --verbose

Tip: nohup pyfs-matrix collect --watch > logs/collect.out 2>&1 &

5. POST-PROCESS (NO SOLVER, NO SEAT)

post [matrix] rebuild polars, sections and plot tables under post/<matrix>; no matrix = every matrix in the manifest. Old products are archived in post/<matrix>/archive/

- sims IDS only these simulations' products, in place (super files kept)
- runs NAME read another manifest (e.g. a rebuilt one)
- from-sims ignore runs.json, build records in memory from sims/ (runs made outside the package); products go to post/<matrix>@sims/
- steps-per-revolution N with --from-sims: for a LAST_REVS_AVG window
- additional-pproc reopen the final .fsm of rows with ADDITIONAL_PPROC (no solve, one launch per point) and extract that pproc; 26.124 only. With it: --fs-version, --fs-exe, --local, --recipe
- diagnostics print recorded post diagnostics as Markdown, change nothing
- strict exit 3 when a product was skipped by design (e.g. polar under sideslip)
- check-frozen refuse instead of warn (see collect)
- force-overwrite DESTROY products instead of archiving (asks; --yes answers it)
- workspace, --pproc-warnings, --verbose

6. SYNC WORKSPACES (E.G. CLUSTER TO WINDOWS)

sync {runs|post|fsm|all} run on the MAIN workspace; peers in inputs/sync-workspaces.toml; levels cumulative, runs < post < fsm < all. Nothing in main is deleted; inputs/ untouched but a declared matrix

- apply change files (default: preview)
- from NAME one peer (default: every non-main one)
- prefer-other on a record conflict take the peer's (default: main wins)
- overwrite on a file conflict archive main's copy and take the peer's
- restore with --apply: rebuild records of sims/ folders no record carries
- include-archives also bring archive folders (skipped by default)
- workspace, --runs NAME

7. RECOVER AND MARK RECORDS

restore {runs|storage|products|plan|additional} restore a file from archive/

- apply (default: preview) --stamp STAMP archive to restore from
- matrix STEM for products and plan, when several matrices have archives
- workspace

rebuild rebuild run records from the folders under sims/

- apply (default: preview) --out NAME write records to another manifest (compare before replacing)
- all-sims every simulation folder, recorded or not --sims 4001,2009 only these
- matrix FILE the matrix the simulations ran from
- build-alias BUILD=ALIAS scheduler name of a build (repeatable), e.g. 26.124=26.12
- inputs-from FOLDER another origin's inputs/ for inputs changed after the run
- workspace

mark-failed every record of the named simulations becomes FAILED_MARKED, its old status kept under marked; runs.json archived first. Not readable before 0.33.0

- sims 2006,2007 the simulations --reason TEXT why, recorded as given
- apply (default: preview) --workspace

8. DISK SPACE (EVERY CALL LOGGED IN storage_management.json)

space-in-use sizes by top folder, by sims/sim_*, by extension; changes nothing

- top N rows per grouping (default 15) --workspace

free-space <m-id> run storage recipe inputs/management/m<id>.toml: prune_step_exports, compact_sims (zip, restored automatically), delete_extensions, post_archives

- apply (default: preview) --runs NAME --workspace
- list every path each step touches, with its size and what happens to it

delete-sims <ids> delete simulations, their products and records, e.g. 4001,2009; a note row remains

- apply (default: preview) --runs NAME --workspace
- force also a simulation whose run is still SUBMITTED (refused without it)
- matrix-products {points-only|regenerate} for a matrix product that also holds the deleted points: leave it (marked stale) or rerun its post without them

Before a licensed run: plan with --cost; read the warnings; inventory --clean a geometry that carries saved actions. Exit codes of post: 0 done; 2 refused, nothing written; 3 with --strict, a product skipped by design. Trace: runs.json, plan.json, post/, logs/.